Introduction to Solar Energy

Solar energy is energy from sunlight that can be converted into electricity or heat. Two main technologies are used today: photovoltaic systems, which convert light directly into electricity, and solar thermal systems, which use sunlight to heat a fluid.

How photovoltaic cells work. A photovoltaic cell is usually made from silicon, a semiconductor. When photons strike the cell, they transfer energy to electrons, creating an electric current. This process is called the photovoltaic effect. Individual cells are connected together into panels, also called modules.

Panel Efficiency and Inverters

Efficiency describes the share of incoming sunlight that a panel converts into electricity. Typical commercial silicon panels convert roughly 18 to 23 percent of sunlight into electricity. Monocrystalline panels are generally more efficient than polycrystalline panels, but they usually cost more to produce.

Solar panels produce direct current, while homes and the electricity grid use alternating current. An inverter converts direct current into alternating current. String inverters serve a whole row of panels, while microinverters are attached to individual panels so that shading on one panel affects the output of the system less.

Panel output decreases as temperature rises, so panels on very hot roofs can produce less electricity than the same panels in cooler but sunny conditions.

Storage and Limitations

Because the sun does not shine at night, many systems include battery storage. Lithium-ion batteries are the most common choice for home storage because of their high energy density and falling prices.

Solar output varies with weather, season and location. Panels in the northern hemisphere usually face south to capture the most sunlight. The variability of solar power is one reason why electricity grids combine it with other sources and storage.

Maintenance requirements are modest: panels have no moving parts, and occasional cleaning plus an inspection of wiring and inverters is usually sufficient. Many manufacturers guarantee that panels will still deliver about 80 percent of their original output after 25 years.